

S&P 500 Synthetic Index Replication via Sparse Portfolio Construction

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1. Problem & Data

Replicate the S&P 500 daily return profile using $k \leq 50$ stocks. Daily OHLCV data spans Jan 2020–Jan 2026; 559 usable constituents after filtering (<10% missing).

Split	Period	Days	Role
Train	Jan 2020–Jun 2024	1,129	Fit selection & weights
Validation	Jul 2024–Jun 2025	250	3 walk-forward folds
Pre-holdout	Jan 2020–Jun 2025	1,379	Final re-fit window
Holdout	Jul–Dec 2025	128	Blind evaluation

2. Methodology

(A) **LASSO**: Regress r_b on $\mathbf{X}$ with ℓ_1 penalty:

$$\hat{\mathbf{w}} = \arg \min_{\mathbf{w}} \|\mathbf{r}_b - \mathbf{X}\mathbf{w}\|_2^2 + \lambda \|\mathbf{w}\|_1$$

λ via LassoCV with 5-fold TimeSeriesSplit. Top- k by $|\hat{w}_i|$ selected, with the help of ℓ_1 norm's diamond shaped constraints.

(B) **Denoising Autoencoder**: Neural Autoencoder(AE) learns a low-dim latent encoding of returns with injected noise ($\sigma = 0.04$). Architecture: In-

put $\rightarrow$ Dense(96) $\rightarrow$ Drop(0.20) $\rightarrow$ Dense(32) $\rightarrow$ Latent(d) $\rightarrow$ mirror,
 $d = \min(12, N/10)$.

- **Latent**: Importance = $|\mathbf{W}_1||\mathbf{W}_2||\mathbf{w}_3^*|$ to best latent dimension
- **Communality**: low reconstruction MSE = high market co-movement

(C) **Markowitz Minimum Tracking Error (MPT)**:

$$\min_{\mathbf{w}} \mathbf{w}^\top \hat{\Sigma}_\epsilon \mathbf{w} + \mu \|\mathbf{w}\|_2^2 \text{ s.t. } \mathbf{1}^\top \mathbf{w} = 1, 0 \leq w_i \leq 8\%$$

$\hat{\Sigma}_\epsilon = \text{Cov}(r_i - r_b)$, so $\mathbf{w}^\top \hat{\Sigma}_\epsilon \mathbf{w} = \text{Var}(r_p - r_b)$. CVXPY/OSQP; $\mu = 10^{-3}$.

(D) **HRP** : Distance $d_{ij} = \sqrt{(1 - \rho_{ij})/2}$, single-linkage clustering, quasi-diagonalisation, recursive IVP bisection:

$$\alpha = 1 - \frac{\hat{V}_1}{\hat{V}_1 + \hat{V}_2}$$

No matrix inversion — robust to estimation error.

Model selection: Rolling CV chooses k per method (min CV TE); sparsity-tolerance rule ($\pm 0.30\%$) prefers smaller k . Final weights re-fit on full pre-holdout window.

3. Results

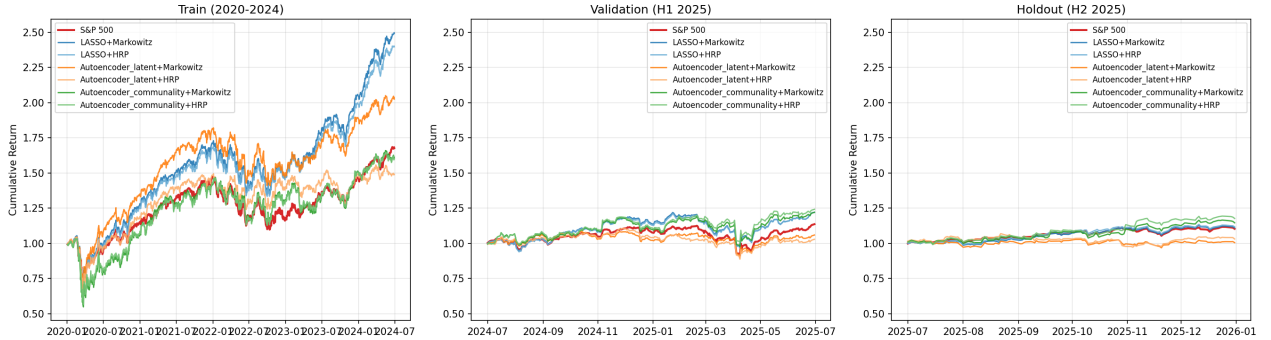


Figure 1: Cumulative returns (train / validation / holdout). LASSO portfolios track the S&P 500 closely. AE latent variants drift negatively in H2 2025. AE communality variants outperform with high TE.

Performance Metrics Summary

Method	Weighting	Split	Selected k	Ann TE (%)	Ann IR	Corr	R ²	Ann Active(%)
Autoencoder_communality	HRP	Holdout	38	12.0005	1.0954	0.5711	-0.2384	13.145
Autoencoder_communality	HRP	Train	38	13.7266	0.0418	0.8815	0.6107	0.5735
Autoencoder_communality	HRP	Validation	38	10.0366	0.9309	0.8862	0.745	9.3428
Autoencoder_communality	Markowitz	Holdout	46	12.2999	0.6553	0.5914	-0.297	8.0603
Autoencoder_communality	Markowitz	Train	46	15.1675	0.0775	0.8724	0.5247	1.1758
Autoencoder_communality	Markowitz	Validation	46	10.243	0.7716	0.8962	0.7347	7.9031
Autoencoder_latent	HRP	Holdout	33	11.7235	-1.1191	0.5145	-0.1822	-13.1197
Autoencoder_latent	HRP	Train	33	7.8054	-0.3152	0.9397	0.8741	-2.4602
Autoencoder_latent	HRP	Validation	33	10.5018	-0.9939	0.8502	0.7207	-10.4376
Autoencoder_latent	Markowitz	Holdout	34	9.6523	-1.8869	0.6435	0.1913	-18.2127
Autoencoder_latent	Markowitz	Train	34	5.4769	0.7789	0.9693	0.9379	4.2657
Autoencoder_latent	Markowitz	Validation	34	8.0412	-0.9122	0.915	0.8364	-7.3355
LASSO	HRP	Holdout	39	4.7838	0.5472	0.9071	0.8039	2.6175
LASSO	HRP	Train	39	4.2636	1.9077	0.9822	0.9619	8.1337
LASSO	HRP	Validation	39	4.6534	1.6134	0.9773	0.9448	7.5079
LASSO	Markowitz	Holdout	40	4.2709	0.3242	0.925	0.8438	1.3848
LASSO	Markowitz	Train	40	3.8421	2.3625	0.9863	0.9688	9.077
LASSO	Markowitz	Validation	40	4.3088	1.7516	0.9813	0.9526	7.5471

Figure 2: Full metrics. **Best: LASSO + Markowitz** ($k=40$, holdout TE = 4.27%, Corr = 0.925). IR = Ann. Active / Ann. TE.

3.1 Sparsity vs. Tracking Error(TE)

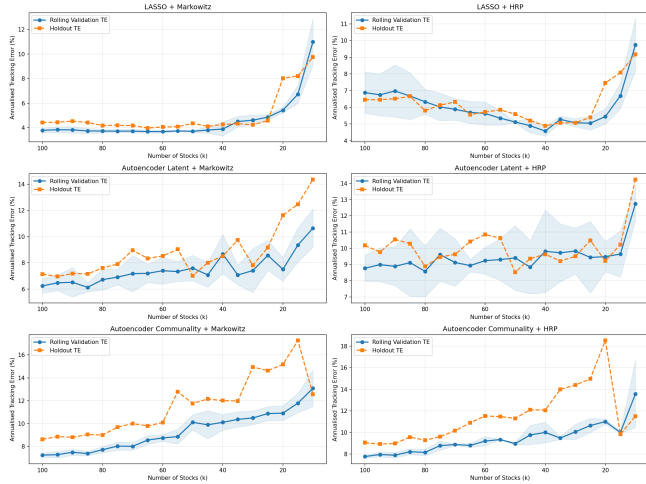


Figure 3: OOS TE (rolling CV) and holdout TE vs. k from 100 to 10. LASSO + MVO < 5% for $k \geq 35$; AE methods degrade sharply below $k=30$.

- **LASSO**: CV TE elbow near $k=30$; $\approx 4\%$ stable for $k \in [35, 100]$
- **Autoencoder Latent**: TE 6–8%, spikes at $k < 25$
- **Autoencoder Communality**: highest baseline ($\approx 8\text{--}10\%$); reconstruction loss $\neq$ TE minimisation
- Holdout TE tracks CV TE for LASSO — no fold leakage

3.2 Information Ratio(IR)

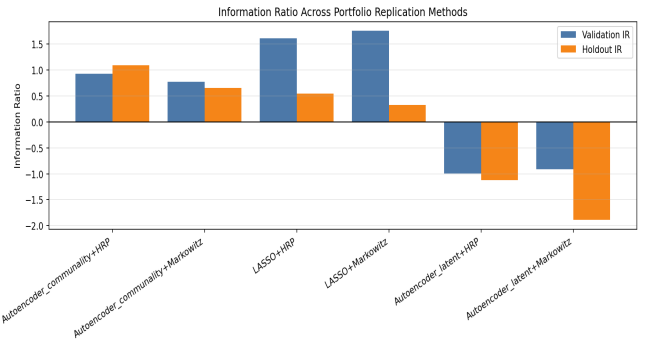


Figure 4: Validation and holdout IR. For pure replication, $IR \approx 0$. LASSO holdout IR: 0.32 (MVO), 0.55 (HRP). AE latent: -1.12 to -1.89 . AE communality: $+0.65\text{--}1.10$ but $TE > 12\%$.

- **LASSO + MPT**: IR near-zero, tightest tracking — ideal replication
- **LASSO + HRP**: slightly higher IR from diversified weights
- **Autoencoder Latent**: large negative IR — systematic drag in H2 2025
- **Autoencoder Communality**: high IR driven by active risk, not tracking quality

3.3 Sector Drift Analysis (LASSO + Markowitz, $k=40$)

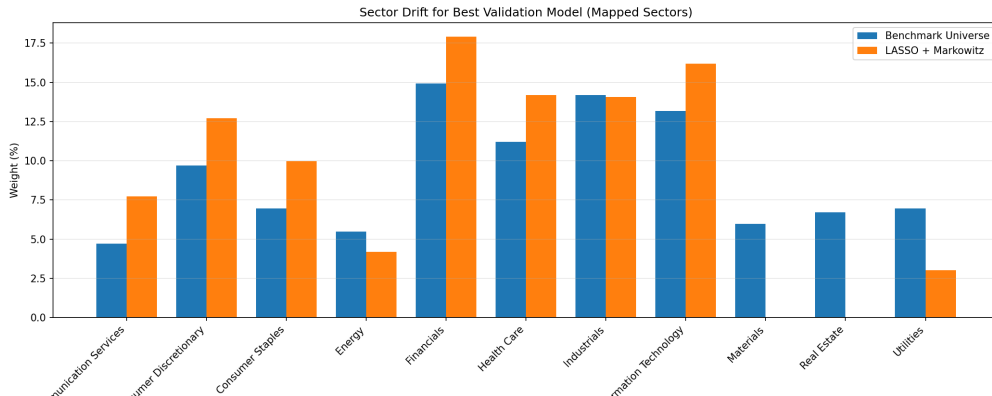


Figure 5: Portfolio vs. equal-weight S&P 500 universe proxy (11 GICS sectors). Max absolute drift < 7pp; all sectors retained. **IT** and **Financials** overweighted by $\sim 3\text{pp}$ (large-cap, high-index-correlation stocks); **Materials**, **Real Estate**, **Utilities** underweighted by $\sim 4\text{pp}$ (low-beta, low cap-weighted correlation).

4. Model Comparison Summary

Method	Weighting	k	Holdout TE (%)	Holdout IR	Holdout Corr	CV TE (%)
LASSO	Markowitz	40	4.27	0.32	0.925	3.90
LASSO	HRP	39	4.78	0.55	0.907	4.65
AE (latent)	Markowitz	34	9.65	-1.89	0.644	6.23
AE (latent)	HRP	33	11.72	-1.12	0.514	7.78
AE (communality)	Markowitz	46	12.30	0.66	0.591	9.64
AE (communality)	HRP	38	12.00	1.10	0.571	9.62

5. Conclusions

LASSO + Markowitz ($k=40$) is the best model: holdout TE = 4.27%, Corr = 0.925, IR = 0.32 — comfortably within the <5% passive-replication target. LASSO's ℓ_1 penalty is a principled heuristic which is balanced by Markowitz min-TE weighting optimally exploiting the selected subset. Autoencoder selection, while theoretically appealing for capturing nonlinear factor structure, does not outperform LASSO: both latent and communality versions are only weakly aligned with $\min \text{Var}(r_p - r_b)$, yielding $TE > 9\%$ across all AE variants.